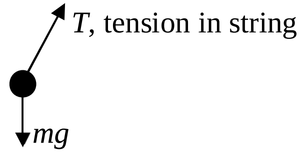
3 **(a)** $T \sin\theta = ma$

$T \cos\theta = mg$



C1

$\tan\theta = \dfrac{a}{g}$

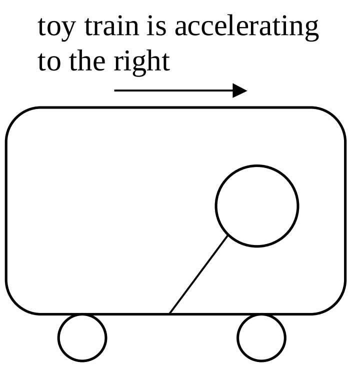
$\theta = 14°$ A1

(b) more air is packed at the back in the accelerating train B1

Therefore higher pressure at back than at front of train B1

Force due to pressure difference pushes the helium balloon forward. B1

Helium (lighter than air) balloon leans forward to region of lower pressure.



(c) **(i)** Friction A1

(ii) $F = \frac{\Delta(mv)}{\Delta t} = 60 \times 2.0$ C1

$= 120$ N A1

(iii) $P = Fv = 120 \times 2.0 = 240\,\text{W}$ A1

(iv) $K = \frac{1}{2}\frac{mv^2}{\Delta t} = \frac{1}{2} \times 60 \times 2^2 = 120\,\text{W}$ A1

(v) Friction between sand and belt
Kinetic energy is lost/ dissipated as <u>internal energy/</u>
<u>Heat in the sand and conveyor belt</u>. A1

4 **(a)** Molecule has component of velocity in three directions